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Reverse Engineering Course

Static Analysis

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Lesson 1 – Assembly to C Code

Riddle1

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- http://data.cyber.org.il/reversing/riddles.rar
- Convert riddle1.exe to C

IDA- Basic

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- Interactive Dis-Assembly
- Basic operations:
  - Sections
  - Memory / Graph views
  - Strings view
  - Show machine code
  - Click / Esc
  - Change names
  - Insert comments
  - Convert string -> data -> string

Reversing Techniques

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- Focus on the important things- attempting to figure every line is wasteful
- Disregard compiler stuff
- Identify signatures of functions

Canary

- Compilers add instructions to verify the stack is untouched
  - Push value and pop it before return
  - Make sure ESP untouched
- Identify by finding matching patterns at func start/end

```
mov     eax, ___security_cookie
xor     eax, ebp
mov     [ebp+var_4], eax
```

```
mov     ecx, [ebp+var_4]
xor     ecx, ebp
call    j_@__security_check_cookie@4
```

Function Signatures

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- Args – passed to the func
- Vars – locals
- Who clears the stack?
- Increase our knowledge over time

Practice

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- Convert riddle2.exe to C

Wonderland

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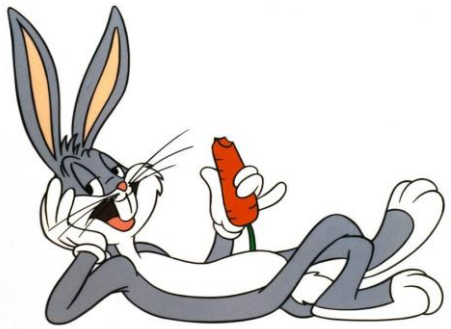
http://data.cyber.org.il/reversing/Wonderland.exe

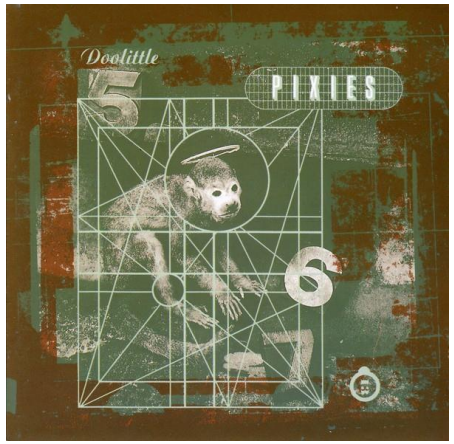
Credit: Major Secret

Practice

- https://data.cyber.org.il/reversing/carrot.zip
  - Find what it does without running and run only afterwards to make sure



- https://data.cyber.org.il/reversing/pixies.zip
  - Find what it does without running and run only afterwards to make sure



Practice

- https://data.cyber.org.il/reversing/vegas.zip
  - Find the password (you may run the executable)
- https://data.cyber.org.il/reversing/foolya.exe
  - Find the password (you may run the executable)